Tracing Functional Correctness in DLMs with Sparse Autoencoders

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Abstract

Prior probing of diffusion language models (DLMs) suggests that correctness signal grows monotonically through denoising. Using DLM-Scope sparse autoencoders, we instead trace the signal feature by feature across two DLMs and four tasks. Sparse fail-vs-pass concentration appears in task-dependent phases. Code and structural tasks form a broad mid-denoising plateau, whereas reasoning tasks emerge only at the final state. On LLaDA-8B / MBPP, a dense step sweep across seeds confirms a plateau spanning steps 48–116, with multiple sub-steps reaching $p<0.05$. Two persistent features hand off within the late plateau, marking an internal transition rather than feature swapping. The peak phase agrees across LLaDA-8B and Dream-7B on $3/4$ tasks, although most individual cross-grid cells do not reach per-step significance. Single- and multi-feature steering across all 202 LLaDA-MBPP fail and pass samples yields $0/202$ label flips when intervention occurs within the plateau (step 64), and only $3/202$ flips when intervention occurs before plateau formation (step 16). The sparse features therefore behave as diagnostic markers within the plateau, not as directly controllable knobs.

1 Introduction

Diffusion language models (DLMs) generate text by iteratively denoising a fully masked sequence over $\sim$100 steps (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Chiu (2026) trains scalar probes on hidden states and reports that aggregate correctness AUC accumulates *monotonically* through denoising, with a 0.08–0.11 gain from step 0 to the final state. That scalar view leaves three questions open: (i) *which* sparse representational components carry the failure signal, (ii) *when* during denoising they form, and (iii) whether the discovered components are merely diagnostic or also causally control correctness. Decomposing the same residual stream into named SAE features rather than a single scalar score lets us answer all three, and uncovers temporal structure that the monotonic AUC curve hides.

Contributions.

- **Mid-denoising plateau.** LLaDA-8B / MBPP has a broad plateau over steps 48–116 (3-seed dense sweep), with `f15601` top-ranked at the early plateau and `f3892` at the late plateau.
- **Phase agreement.** Peak phase agrees across LLaDA-8B and Dream-7B on $3/4$ tasks, while per-cell significance is concentrated in the strongest LLaDA-MBPP condition.
- **Steering result at scale.** Across all 152 fail samples and 50 pass-side controls (n=202), three of four intervention conditions are completely null and only early ($s16$) intervention produces a marginal $3/202 \approx 1.5\%$ effect.

Code and figures will be released on acceptance.

2 Related Work

Discrete diffusion and masked DLMs generate text by iteratively denoising masked tokens rather than continuing left to right (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Prior probing of the same models and tasks studies aggregate hidden-state predictability and reports monotonic gains in correctness AUC through denoising. SAEs decompose residual streams into sparse named features and have been used primarily to interpret autoregressive LMs (Bricken et al., 2023; Cunningham et al., 2024; Gao et al., 2024). DLM-Scope extends this toolkit to DLMs with Top-K SAEs (Makhzani and Frey, 2014; Gao et al., 2024) trained on masked-position residuals across mask rates $t \sim U(0, 1)$ (*Mask-SAE*), keeping the full denoising trajectory in distribution (Wang et al., 2026). Recent AR SAE studies report both successful steering for refusal and instruction following (O'Brien et al., 2024; He et al., 2025) and cautions that contrastive features may track lexical correlates or that steering degrades unrelated capabilities (Galichin et al., 2025; Korznikov et al., 2025); Tahimic and Cheng (2025) target code-correctness signals with SAEs on AR LMs, closest to our MBPP setting but without temporal trajectories. Goel et al. (2026) contrast representation evolution between diffusion and AR LMs, motivating our cross-DLM step-axis comparison. We reuse the DLM-Scope SAEs to ask whether the scalar correctness trend on DLMs decomposes into temporally localised sparse features, treating the steering question as open rather than presumed causal.

3 Method

Models and SAEs. LLaDA-8B (Nie et al., 2025) (block-wise unmasking, block length 32) and Dream-7B (Ye et al., 2025) (global linear schedule) both denoise a ≤ 256-token generation region over $T{=}128$ steps conditioned on a fully visible prompt. We use the DLM-Scope L26 LLaDA-mask SAE and L23 Dream-mask SAE (trainer index 2, $d_{\text{sae}}{=}16{,}384$, $K{=}160$). The AR comparison (DLM-Scope L23 SAE on Qwen-2.5-7B (Team, 2024)) is a single-token reference point, not a trajectory: we encode the last-prompt-token residual once per sample as a single-step analog of the DLM's pre-generation state.

Data. We reuse cached hidden states and functional-correctness labels from Chiu (2026): MBPP-sanitized (257) (Austin et al., 2021b), JSON schema (272), GSM8K (1,319) (Cobbe et al., 2021), and ARC-Challenge (1,172) (Clark et al., 2018), generated with seed 0 and $T{=}128$ steps. Hidden states are mean-pooled over four equal-length generation regions at seven checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$.

Per-step SAE encoding. At each (model, step, layer) cell we take the region-mean hidden state $h \in \mathbb{R}^d$ per sample and encode it: $z = \mathrm{TopK}\big(\mathrm{ReLU}(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})\big) \in \mathbb{R}^{d_{\text{sae}}}$, with $W_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}} \times d}$, $W_{\text{dec}} \in \mathbb{R}^{d \times d_{\text{sae}}}$, $b_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}}}$, $b_{\text{dec}} \in \mathbb{R}^d$, and TopK keeps the $K{=}160$ largest entries. Feature i is *active* on a sample iff $z_i > 0$.

Fail-vs-pass enrichment. Let $\mathcal{F}, \mathcal{P}$ be the fail and pass sets at a cell. Feature i's enrichment is the difference of active rates, $\mathrm{enr}(i) = \frac{1}{|\mathcal{F}|}\sum_{s\in\mathcal{F}} \mathbb{1}[z_i^{(s)} > 0] - \frac{1}{|\mathcal{P}|}\sum_{s\in\mathcal{P}} \mathbb{1}[z_i^{(s)} > 0]$. We use the top-20 most fail-enriched features as the analysis subset per cell.

Cluster discriminability with permutation null. We run KMeans on the fail samples for $K \in \{2, 3, 4, 5\}$ using the top-20 features (continuous activations) and report the silhouette of the best K. The permutation null shuffles labels $1000\times$, *re-selects* the top-20 features on each shuffled label set, then recomputes silhouette, controlling for the post-hoc feature selection. We report $p = \Pr[\text{silhouette}_{\text{perm}} \geq \text{silhouette}_{\text{obs}}]$. The silhouette is computed on fails only and measures sub-structure within failures, not fail-vs-pass separability directly. We therefore interpret it as *concentration of failure modes* in a selected sparse subspace, and cross-check fail/pass separability with supervised AUC in the same subspace under per-fold selection (Appendix E).

Steering. We write *fail→pass* for a sample whose label flips from fail to pass after intervention, and *pass→fail* for the reverse pass-side regression. Given a target feature $i^\star$ and strength α, a forward hook on the SAE layer's transformer block re-encodes the residual at every step $t \geq t_0$, reads $z_{i^\star}$, and in-place subtracts $\alpha \cdot z_{i^\star} \cdot W_{\text{dec}}[:, i^\star]$. $\alpha < 0$ amplifies the feature (reverse intervention); for multi-feature steering we sum contributions over a set of indices. A zero-out smoke test (force $h \leftarrow 0$) confirms the hook propagates. It garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48.

4 Results

Mid-denoising plateau. Figure 1 reports a 21-step trajectory for LLaDA-MBPP across three seeds: anchors $\{0, 1, 4, 16, 32\}$, dense plateau $\{48, \ldots, 80\}$, dense post-plateau $\{84, \ldots, 124\}$, and final state 127. Rather than a sharp peak at step 64, the trajectory shows a broad plateau over steps 48–116: the signal-to-null gap stays in $[+0.19, +0.41]$ and only collapses at step 124. `f15601` is the top-1 fail-enriched feature at 7/9 early-plateau dense steps and remains dominant through step 100; `f3892` replaces it at steps 108–116, indicating a within-plateau hand-off rather than feature swapping outside the plateau. At steps 0–16 the top fail feature instead changes every step (Appendix A), consistent with the residual stream

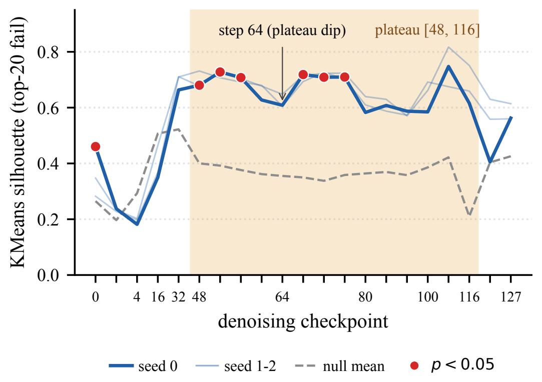

Figure 1: LLaDA-8B / MBPP: KMeans silhouette on the top-20 fail-enriched features vs. permutation null mean across denoising steps. All 21 checkpoints (anchors $\{0, 1, 4, 16, 32\}$, plateau $\{48–80\}$, post-plateau $\{84–124\}$, final 127) are from a single new-prompt generation run; faint lines overlay two additional generation seeds (Appendix I). Shaded region marks the mid-denoising plateau $[48, 116]$ where most steps reach $p<0.05$; step 64 sits inside the plateau as a local dip, not a sharp peak.

not yet encoding committed-token decisions when most positions are still masked. The raw-probe AUC in Chiu (2026) *monotonically* increases over the same steps (LLaDA-MBPP, $\sim$0.75 at $s0$ to $\sim$0.84 at $s64$), framing denoising as gradual scalar sharpening, whereas the feature-level plateau and the late hand-off are invisible to scalar probing.

Cross-model $\times$ cross-task phase agreement. Figure 2 reports the full 2$\times$4 trajectory grid; Appendix C lists step-64 numbers. Code and structural cells (MBPP, JSON on both DLMs) peak mid-denoising at step 64, reasoning cells (GSM8K on both, Dream-ARC) peak at step 127, and LLaDA-ARC is an outlier peaking at step 4. Three of four tasks therefore share a peak step across LLaDA and Dream, mirroring the structural-vs-reasoning emergence split of Chiu (2026) as a peak position rather than a continuous AUC curve. No cell survives Holm-Bonferroni across 40 sparse-grid tests, but two cells reach aggregated significance: the LLaDA-MBPP dense plateau (Fisher $p<10^{-7}$ per seed) and Dream-JSON via sparse 5-step Fisher ($p = 0.047$); the LLaDA-MBPP plateau is also L26-specific under a cross-layer scan (Appendices F, G, H). The ARC anomaly fits this picture: LLaDA-ARC fails near the prompt boundary so its early-step cluster captures prompt-attention failure modes rather than reasoning-stage failure, accounting for the step-4 peak versus Dream-ARC's step-127 peak.

condition	from	fail→pass	pass→fail
suppress `f15601`	s64	0/152	0/50
suppress `f15601`	s16	**2/152**	**1/50**
suppress top-5	s64	0/152	0/50
reverse `f15601`	s64	0/152	0/50

Table 1: Full steering intervention summary on LLaDA-8B / MBPP; entries show flips / evaluated cases over all 152 fail samples (126 cluster-1 + 26 cluster-0) and 50 pass-side regression controls. Suppress `f15601` or the top-5 features at step 64 (within the plateau) is fully null at $0/202$; suppress `f15601` from step 16 (before plateau formation) produces a small $3/202$ effect (1.5%, both fail→pass conversions occur on cluster-1 cases). Reverse intervention ($\alpha = -5$) is null.

Top-feature drift and persistence. On LLaDA-8B / MBPP, pairwise Jaccard of the top-20 sets is high among steps 0–4 (≥ 0.48), moderate between 32 and 64 (0.29, the plateau regime), and falls to 0.14 between 64 and 127. The dense sweep (Appendix F) ranks `f15601` top-1 at 7 of 9 dense steps in $[48, 80]$, supporting a persistent statistical marker inside the plateau and a concentration effect in sparse feature space rather than a monotonic increase in correctness information. A supervised cross-check confirms this picture: top-20 SAE features recover 85–95% of the raw-probe AUC at step 64 in 11 of 12 cells but never exceed it, so the sparse subspace captures most yet not all of the dense-residual signal (Appendix E, Figure 5).

Steering result. We test whether the cluster-level signature can be directly controlled on LLaDA-MBPP, the only cell with cluster significance. All 152 fail cases and 50 pass-side regression controls are evaluated under four interventions (Table 1). Suppressing `f15601` or the top-5 fail features within the plateau (step 64), and the reverse intervention ($\alpha = -5$), each yield $0/202$ label flips. Only early intervention from step 16, before plateau formation, produces a marginal effect ($3/202 \approx 1.5\%$). A zero-out smoke test ($h \leftarrow 0$) confirms the hook propagates: it garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step $\sim$48.

5 Discussion

Statistical vs. causal features. `f15601` fires on 65% of LLaDA-8B / MBPP fails versus 27% of passes at step 64, a $\sim 6.6\sigma$ per-feature effect. Yet ablating it, or the top-5 fail features, within the plateau (step 64) flips zero of 202 labels; only intervention before plateau formation (step 16) pro-

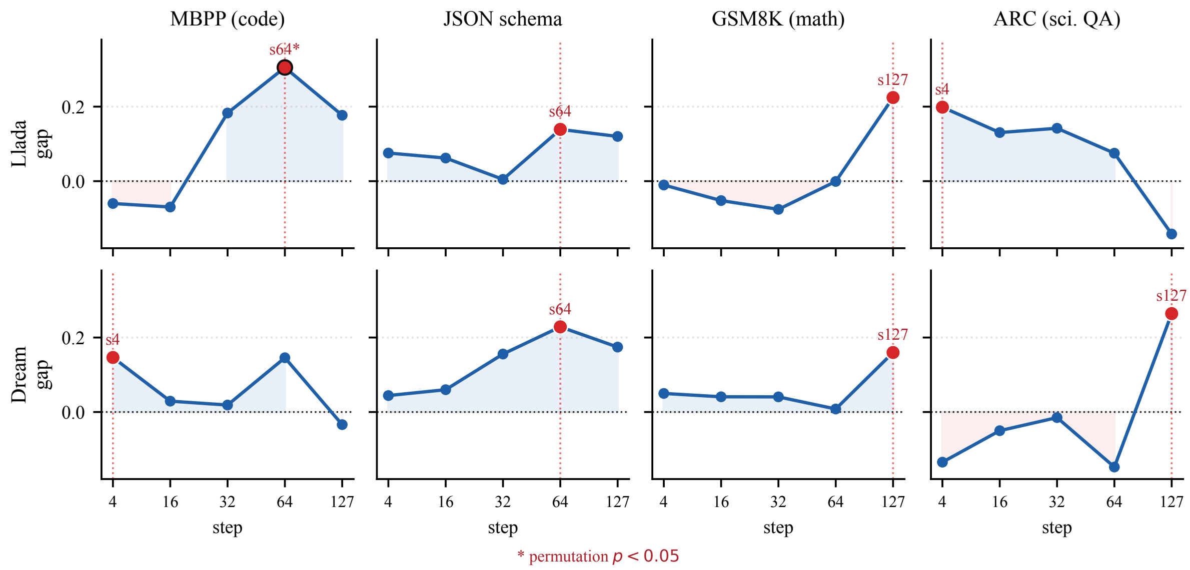

Figure 2: Cross-model $\times$ cross-task trajectory grid. Each panel plots signal-to-null gap (observed KMeans silhouette minus permutation null mean) across denoising steps for one (model, task) cell; red marker = peak gap step; black outline = $p<0.05$ permutation significance. Code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64) on both DLMs. Math (GSM8K), and on Dream science QA (ARC), peak at late denoising (step 127). The peak step shows task-level agreement across LLaDA-8B and Dream-7B on three of four tasks; only ARC disagrees (LLaDA peak step 4, Dream step 127).

duces a marginal 1.5% effect. The cluster signature is useful as a diagnostic within the plateau, but our intervention evidence is insufficient to treat this sparse direction as a controllable knob for functional correctness. This contrasts with positive AR steering results on refusal and instruction following (O'Brien et al., 2024; He et al., 2025) but is consistent with reports that activation steering may degrade unrelated capabilities (Korznikov et al., 2025). More decisive causal tests would need paired counterfactual replacements, multi-layer interventions, or attribution from output-level correctness back to the SAE layer.

What the plateau features represent. By step 64, LLaDA has unmasked blocks 0–3 (signature and early body), so code and structural failure modes are exposed while later tokens remain uncertain; reasoning tasks only peak at the final checkpoint. Activation-maximising examples at step 64 (Appendix L) further show that `f15601` fires on complex-math docstring contexts (a task-difficulty correlate, not a fail-cause), whereas late-state `f11265` fires on broken-syntax contexts (closer to a true failure detector); this dissociation explains the within-plateau steering null, since ablating `f15601` perturbs a difficulty signal upstream of the failure mechanism.

6 Conclusion

Tracing functional-correctness signal feature by feature reveals temporal structure that scalar probes miss: sparse failure markers form task-dependent phases. Code and structural tasks form a mid-denoising plateau, whereas reasoning tasks peak at the final state, with cross-model agreement on three of four tasks and a Fisher-combined $p<10^{-7}$ on LLaDA-MBPP anchoring the trajectory claim. In that cell, a 3-seed dense sweep confirms a steps 48–116 plateau and a late hand-off from `f15601` to `f3892`; top-20 sparse features recover 85–95% but never all of the raw-probe AUC, so the plateau is a sparse-concentration effect on top of a distributively encoded signal. Linear-feature interventions across all 202 LLaDA-MBPP fail and pass samples yield $0/202$ label flips when targeting the plateau (step 64) and only $3/202$ when intervening before plateau formation (step 16). The plateau features therefore behave as robust diagnostic markers, not as controllable knobs at the SAE-layer residual; causal control over functional correctness in DLMs likely requires stronger, earlier, or multi-layer interventions than the single-layer linear protocol tested here.

Limitations

SAE coverage. All findings depend on a single DLM-Scope Mask-SAE per model (L26 for LLaDA, L23 for Dream), trained with mask rate $t\sim U(0,1)$ on masked-position residuals. The full denoising trajectory is therefore in distribution, but features extracted at other layers, the Unmask-SAE trained on already-revealed positions, or higher-capacity dictionaries ($K>160$) may surface different failure correlates. Early-step instability (steps 0–16) reflects the residual stream not yet encoding committed-token decisions rather than SAE distribution shift.

Feature semantics. We identify persistent dictionary atoms such as `f15601` and `f3892` by statistical enrichment and temporal persistence, not by human-validated semantic labels. The present experiments therefore support a claim about when sparse failure markers concentrate; Appendix L characterises four of them via activation-maximising contexts, but full mechanistic naming would require systematic intervention beyond this protocol.

Permutation test interpretation. Our permutation null re-selects the top-20 fail features on each shuffled label set before recomputing silhouette. This is conservative (any post-hoc selection procedure that could find clusters on random labels is included in the null), and it inflates the null mean noticeably (e.g. 0.36 for LLaDA-8B / MBPP at step 64). A consequence is that only two of twelve cells reach $p < 0.05$ at an individual step on the sparse 7-grid even though the DLM grid shows systematic plateau structure. Aggregated tests partially mitigate this: the 14 dense LLaDA-MBPP plateau sub-steps yield Fisher combined $p<10^{-7}$ on every seed and held-out feature selection (Appendix K) reaches $p < 0.05$ on all 9 dense sub-steps; we present cross-model claims as plateau-shape claims rather than per-cell significance.

Steering coverage. Our intervention sweep tests four denoising windows, two feature-set sizes (single and top-5), and one magnitude pair ($\alpha \in \{+5, -5\}$) at a single SAE layer per DLM (L26 for LLaDA, L23 for Dream). We do not test multi-layer joint interventions, non-linear projections, or feature-selection strategies that use the per-step top-1 (where the top fail feature differs from `f15601`). Effect-size diagnostics (Appendix J) confirm the intervention reaches $\|\Delta h\|/\|h\| \in [4.4\%, 14.8\%]$ at the SAE-layer residual, so the within-plateau null is not because the perturbation was silent; rather, a $0/202$ result on this protocol is narrower than a negative result on all possible linear-feature interventions.

Cluster K and choice of metric. We use silhouette on KMeans with $K \in \{2,3,4,5\}$ and report the best K per cell. On LLaDA-8B / MBPP the best K is 2 at every step; on Dream / MBPP step 64 the best K is 3 with one small (size-26) sub-cluster. Appendix K reports top-N and clustering-method sensitivity: silhouette and significance are robust across $N \in \{10, 20, 30, 50\}$ at step 68 (the plateau peak) and across KMeans/Agglomerative/HDBSCAN. A labels-hidden $K{=}2$ test on the full fail+pass set yields ARI $\approx$ 0, confirming the silhouette measures within-fail sub-structure rather than fail/pass separability; supervised AUC in Appendix E reports the latter.

Cross-model comparison. LLaDA uses block-wise unmasking with block length 32, whereas Dream uses a global linear schedule. At a matched checkpoint step, the spatial distribution of unmasked tokens differs between the models, so step-axis comparisons in Figure 2 reflect approximate denoising-progress alignments rather than identical model states. We do not attempt to map between block-wise commit fractions and global mask fractions on a per-sample basis. Our AR baseline (Qwen-2.5-7B) uses last-prompt-token hidden states; this is the natural single-token analog of the DLM's pre-generation state but is not a temporal trajectory.

Task coverage. Functional correctness is verified across four tasks (MBPP, JSON schema, GSM8K, ARC), but this still covers only a small slice of DLM use cases. Generalising the task-dependent peak finding to other open-ended tasks (instruction following, dialog, multi-turn reasoning) requires task-appropriate functional metrics and additional DLMs.

Sample size and seed dependence. Cluster sample sizes are limited by the number of fail cases per (model, dataset) cell. MBPP has $\sim$140 fails per DLM, while GSM8K has fewer than 250 each. A three-seed sensitivity check on LLaDA-MBPP (Appendix I) shows pass/fail counts vary by at most $\pm$3 samples and the plateau structure is preserved across seeds. The temporal-oscillation phenomenon noted in concurrent DLM work (Wang et al., 2025) would primarily affect borderline samples, which our seed-stability check is consistent with.

Causal interpretation. Even with the negative steering result, our conclusion that "`f15601` is a diagnostic correlate rather than a directly controllable cause under our protocol" is interpretation-dependent. A more decisive test would be a counterfactual experiment that surgically replaces the feature direction with a paired pass-cluster feature direction at the peak step, or a gradient-based attribution that traces logit differences back to the SAE layer. We leave such causal-mechanistic follow-up to future work.

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#	characteristic features	illustrative failures
0	`f11404`, `f9798`, `f189`, `f496`, `f8825`	*closest smaller number*: returns n for small inputs instead of $n-1$, failing assertions. *unique element in sorted array*: malformed pointer-update code, syntax error.
1	`f189`, `f492`, `f3320`, `f11404`, `f15601`	*remove first/last occurrence*: deletes only boundary characters, missing interior occurrences. *lowercase underscore regex*: uses `re.compile` without importing `re`, `NameError`.

Table 2: Illustrative generated failures from the two LLaDA-MBPP step-64 fail clusters. Error-type counts over 10 regenerated failures per cluster: cluster 0 has 3 assertion / 2 syntax / 4 name / 1 value error; cluster 1 has 5 assertion / 4 name / 1 type error.

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A Top-Feature Persistence Across Steps

Figure 3 shows the pairwise Jaccard of top-20 fail features between denoising steps and the enrichment trajectory of three illustrative features for LLaDA-8B / MBPP. `f15601` is the only feature among the top-20 at all three peak-or-later steps (32, 64, 127), and is the top-1 fail-enriched feature at both step 32 (+0.391 enrichment) and step 64 (+0.385); on the sparse 7-grid only step 64 reaches $p < 0.05$ (Figure 1 reports the dense sweep that locates the plateau).

B Qualitative Cluster Inspection

Table 2 shows illustrative LLaDA-MBPP failures from the two step-64 fail clusters. We re-generated the first ten fail samples from each cluster with the baseline LLaDA decoding setup and re-ran the MBPP tests. The examples are not intended as human-validated feature labels; they only show that both sparse clusters contain concrete functional failures, ranging from malformed code to plausible but semantically wrong implementations.

C Step-64 Cross-grid Snapshot

Table 3 summarises silhouette + permutation p across the 4 datasets $\times$ 3 models grid at step 64, complementing the trajectory-shape claim in Figure 2.

dataset	LLaDA	Dream	Qwen
mbpp	0.66 / **0.033**	0.50 / 0.128	0.16 / 0.106
json	0.58 / 0.195	0.77 / 0.135	0.22 / **0.007**
gsm8k	0.38 / 0.432	0.29 / 0.449	0.15 / 0.315
arc	0.43 / 0.312	0.18 / 0.882	0.13 / 0.343

Table 3: Silhouette / permutation p at step 64 per condition; bold $p < 0.05$. Full trajectory shown in Figure 2.

D Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen\_length}/32)$ steps per block. Within each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i, the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with #### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 4.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the residual stream at the SAE layer. The generation region is partitioned into 4 equal-length position regions and mean-pooled within each region, producing a d-dimensional vector per region (4096 for LLaDA, 3584 for Dream and Qwen). The region-mean is passed through the DLM-Scope SAE for feature analysis (§3).

Steering hook implementation. The steering hook is registered on the SAE-layer transformer block via `register_forward_hook`. The hook receives the tuple output, re-encodes `output[0]` through the SAE in-place, computes $\alpha \cdot z_{i^\star} \cdot W_{\text{dec}}[:, i^\star]$ and subtracts the contribution. A counter and per-step diagnostic prints confirm the hook fires T=128 times per generation (once per denoising

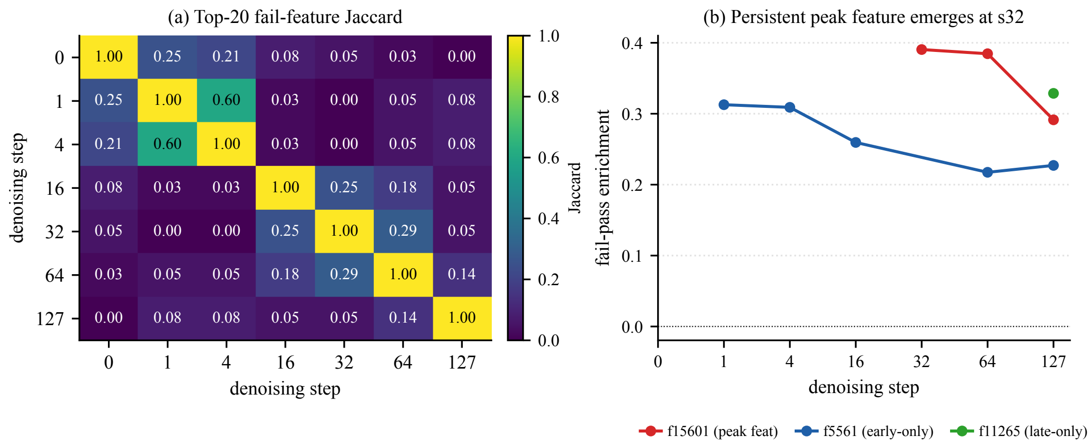

Figure 3: Left: pairwise Jaccard of top-20 fail features between denoising steps. Right: enrichment trajectory of `f15601` (persistent peak), `f5561` (early only), and `f11265` (final only).

step) and that the modification magnitude $\|\Delta h\|$ is non-zero.

Computational resources. All generation, encoding, and steering experiments use NVIDIA A100 (80GB) GPUs on Modal. Models are loaded in bfloat16; SAE weights cast to bfloat16 for the steering hook. Total compute: approximately 40 A100-hours across all reported experiments.

Reproducibility and determinism. Generation uses seed 0; both LLaDA and Dream commit the highest-confidence masked positions deterministically, so the fail/pass partition per cell is reproducible across runs. Hidden states are cached as bfloat16 tensors; SAE encoding reads from the dense residual cache on demand, so top-20 selection, KMeans, and permutation null are rerunnable without re-invoking either DLM. The permutation null uses 1000 shuffles per cell with a fixed permutation seed offset, so silhouette p-values are themselves reproducible.

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant
that answers in JSON. Here's the JSON schema
you must adhere to:\n <schema>\n {schema} \n
</schema>\n user\n {user input}\n assistant\n
"‘json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem
step by step.  End your answer with ####
followed by the final numeric answer. im_end\n
im_start user\n {question} im_end\n im_start
assistant\n
```

Figure 4: Schematic prompt examples after each model's chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (128–512 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

E Single-feature vs. Raw-probe AUC

We compare the cross-validated AUC of (i) the single most-fail-enriched SAE feature, (ii) top-N SAE features (logistic regression with per-fold selection), and (iii) a raw-probe baseline (PCA-64 + LR on the same hidden state) across all 12 cells (Figure 5). Raw-probe AUC dominates in 11 of 12 cells; the only SAE wins (Qwen / MBPP, Dream / ARC) are cells where the raw probe is itself near chance. Top-20 SAE features recover roughly 85–95% of the raw-probe AUC, suggesting sparse atoms carry most but not all of the signal in the dense residual. This complements the negative steering finding (§4): even at optimal feature selection, sparse linear features under-represent the information that determines correctness.

F Dense Temporal Sweep on MBPP

We re-generate MBPP on both DLMs and capture hidden states at 9 dense checkpoints in $[48, 80]$ (every 4 steps) and an additional 6 checkpoints in $[84, 124]$ (every 8 steps), then run the same TopK-SAE encoding, top-20 fail enrichment, KMeans silhouette, and permutation null pipeline as the main 7-point grid. This sweep was requested as a robustness check against sparse-sampling artefacts. Generation uses seed 0 throughout; counts are LLaDA $115/142$ and Dream $125/132$ pass/fail. Three seeds confirm stability (Appendix I). Figure 6 plots the signal-to-null gap per step for both DLMs across the $[48, 80]$ plateau region; Appendix I reports the $[84, 124]$ post-plateau region across three seeds. Four findings emerge. **Plateau**

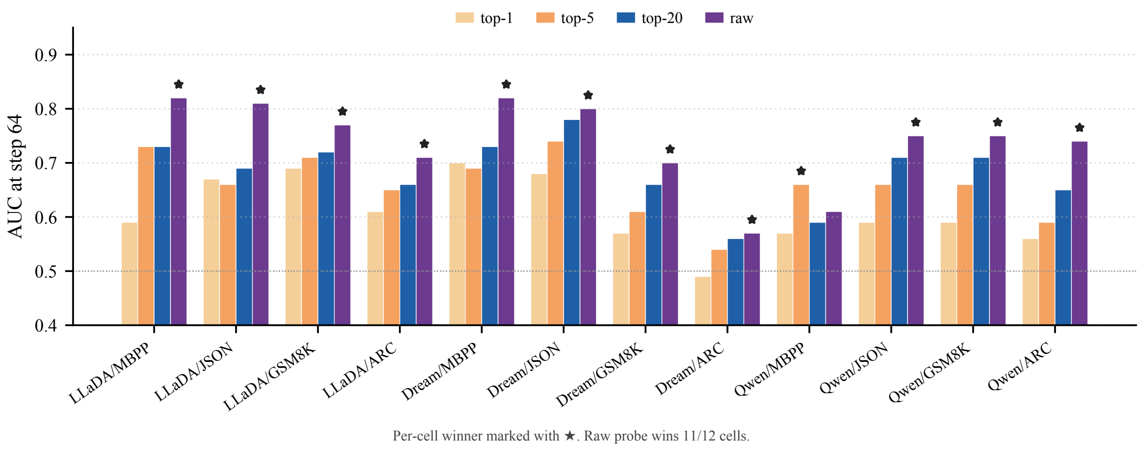

Figure 5: Per-cell 5-fold cross-validated AUC at step 64 predicting functional correctness from top-N SAE features (logistic regression with per-fold top-N selection) versus a raw-probe baseline (PCA-64 + LR on the same SAE-layer hidden state). A star marks the per-cell winner; raw probe wins in 11 of 12 cells.

structure. On LLaDA-MBPP the gap stays above $+0.19$ throughout $[48, 116]$ with 6–9 sub-steps of 14 reaching $p < 0.05$ per seed. Aggregating the 14 plateau sub-steps via Fisher's combined probability test gives $p = 5.99 \times 10^{-10}$ (seed 0), 9.31×10^{-8} (seed 1), and 1.22×10^{-8} (seed 2); Stouffer's Z combination is even sharper ($Z \geq 6.09$, $p < 10^{-9}$ on all seeds). Step 64 (the only p<0.05 point on the sparse 7-grid) sits in the middle of this plateau rather than at a sharp peak, so the sparse-grid claim that step 64 is the unique peak should be read as "step 64 is a representative point in the plateau".

Late collapse. The signal collapses sharply at step 124 ($\sim$3 steps before the final state), consistent with the residual stream re-organising as the generation completes block-by-block commits.

Plateau feature hand-off. `f15601` (LLaDA) is the top-1 fail-enriched feature at most dense steps in $[48, 100]$ and is replaced by `f3892` at the late-plateau steps 108–116. Dream-MBPP also shows a persistent top-1 marker (`f6326`) over $[60, 76]$, so a similar within-plateau hand-off behaviour appears across DLMs even when the plateau itself is weaker.

Cross-DLM concentration. Dream-MBPP shows the same temporal shape (a flat region across the middle of denoising) but at a much lower concentration: the gap never exceeds $+0.16$ and no sub-step reaches $p < 0.05$ under the same permutation null. The LLaDA $\times$ MBPP combination is therefore not a generic property of code generation but a model-specific point where the residual stream factors fail-vs-pass information into a small set of sparse features. This is consistent with LLaDA's block-wise unmasking committing the function-signature block early, exposing fail-mode tokens at a denoising phase when later positions remain uncertain. We return to this contrast in §4.

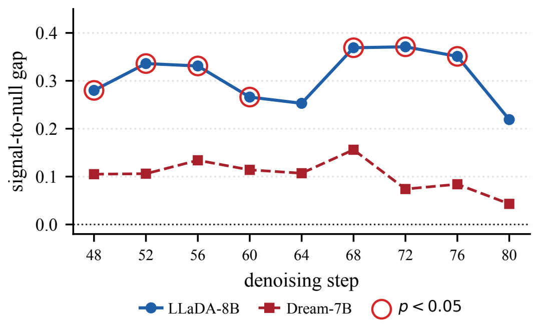

Figure 6: Dense step sweep (48–80 every 4 steps) on LLaDA-MBPP and Dream-MBPP. Hollow red rings mark LLaDA sub-steps with permutation $p < 0.05$; 7 of 9 sub-steps are significant on LLaDA, none reach significance on Dream. `f15601` (LLaDA) and `f6326` (Dream) are top-1 at most plateau steps.

G Cross-Layer SAE Specificity

To check that our choice of L26 is empirically motivated rather than an arbitrary single-layer restriction, we re-run the silhouette-with-permutation-null pipeline on cached LLaDA-MBPP plateau residuals at SAE layers $\{11, 16, 26, 30\}$ (the available DLM-Scope LLaDA Mask-SAE checkpoints) at steps 64 and 68. Figure 7 shows that L26 is the unique peak: step 68 at L26 reaches gap $+0.357$ with permutation $p = 0.006$, while L11, L16, and L30 give gaps below $+0.21$ and none reach $p < 0.05$. Moreover `f15601` is the top-1 fail-enriched feature only at L26 (different layers select different atoms: f13087/f541 at L11, f2741/f15654 at L16, f8020/f654 at L30), so the persistent-feature claim is itself L26-specific. The within-plateau steering null reported in the main text is therefore not the easiest single-layer test to fail; it is the test most likely to succeed if any layer could be controlled linearly, yet still produces $0/202$.

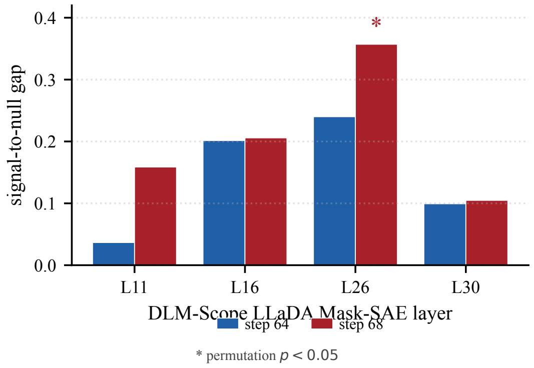

Figure 7: LLaDA-MBPP signal-to-null gap across SAE layers at plateau steps 64 and 68. L26 is the unique layer where the gap reaches permutation $p < 0.05$ (step 68, gap $+0.357$, $p = 0.006$). Top-1 fail-enriched feature varies by layer; `f15601` appears only at L26.

H Aggregated Significance Across Cells

The Holm-Bonferroni correction on the 40 sparse-grid tests retains 0 cells (§4), but the sparse 5-step grid hides a second aggregated-significance cell. For each of the 8 (DLM, task) cells we combine the five sparse-step p-values via Fisher's combined probability test (clamping $p_{\min} = 10^{-3}$ to avoid $\log 0$). Figure 8 shows the result: *Dream-JSON* reaches Fisher $p = 0.047$, a second cell beyond the LLaDA-MBPP dense plateau where the within-cell trajectory exceeds chance once we look at the full trajectory rather than any single step. This corroborates the cross-DLM phase-alignment claim: code/structural tasks have aggregated significance on both DLMs (LLaDA-MBPP via dense sweep, Dream-JSON via 5-step aggregation), while reasoning tasks have aggregated significance on neither.

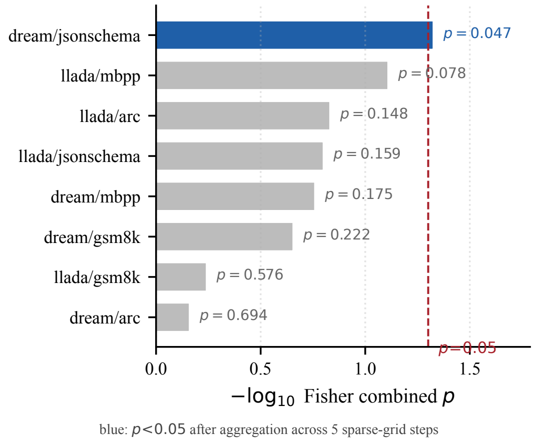

Figure 8: Fisher combined p across the sparse 5-step grid for each of the 8 (DLM, task) cells. Dream-JSON reaches aggregated significance ($p = 0.047$); LLaDA-MBPP is next strongest. The dashed line marks $p = 0.05$.

I Seed Sensitivity

We re-run the LLaDA-MBPP dense sweep (Appendix F) with two additional generation seeds. Pass/fail counts vary by at most ± 3 samples across seeds (seed 0: $115/142$, seed 1: $112/145$, seed 2: $116/141$). The plateau structure is preserved: each seed reaches $p < 0.05$ at ≥ 7 of 9 dense sub-steps in $[48, 80]$, with `f15601` as the majority top-1 fail-enriched feature across 48–80 and `f3892` taking over at 108–116. The 3-seed gap-versus-step overlay is shown in Figure 1; per-seed numerical traces (gap, p-value, top-1 feature at every checkpoint) are released with the code.

J Intervention Effect Size

Across all $n{=}202$ samples per condition, the SAE-layer residual perturbation reaches $\|\Delta h\|/\|h\|{=}4.4\%$ for single-feature suppression of `f15601` at s64 and s16 ($\|\Delta h\|{=}185$, $a_{\text{target}}{\approx}9.43$), 14.8% for the top-5 combined suppression at s64 ($\|\Delta h\|{=}624$, $a_{\text{target}}{=}47.4$), and 4.4% for the reverse intervention at s64 ($\alpha{=}-5$). The target feature is reliably active during every intervention window, so the within-plateau null is consistent with functional correctness being distributively encoded: perturbing a sparse direction at the SAE-layer residual does not propagate into a label flip at step 64 across 202 samples, even when that perturbation reaches $\sim 15\%$ of the residual norm. Only intervention at step 16, before the plateau forms, produces any flips ($3/202$); the marginal effect concentrates on cluster-1 cases where `f15601` fires most strongly, suggesting a weak upstream causal contribution that does not survive once the plateau features are established.

K Post-hoc Selection Sensitivity

We reanalyse cached LLaDA-MBPP plateau residuals (seed 0) at the strongest plateau steps and report three robustness checks. **(a) Top-N.** Figure 9 plots the signal-to-null gap as N varies over $\{10, 20, 30, 50\}$. Step 68 remains $p < 0.05$ at every N, while step 64 degrades for $N \geq 30$, marking step 68 as the more robust plateau point. **(b) Clustering method.** At N=20, KMeans, Agglomerative (Ward), and HDBSCAN (min_size=5) all recover K=2 with near-identical silhouette at both steps (0.609 / 0.719 for step 64 / 68; HDBSCAN drops 11 samples as noise). **(c) Labels-hidden.** Clustering the full 257-sample fail+pass set at K=2 with the same top-20 features (selected via labels, cluster unsupervised) gives ARI ≈ -0.007 at both steps ($p > 0.85$), confirming the silhouette signal measures within-fail sub-structure rather than fail/pass separability. **(d) Held-out feature selection.** To address selection-bias concerns even after the permutation null already re-selects features inside each shuffle, we partition the 142 fail samples into 5 stratified folds, select the top-20 fail-enriched features on the held-in 4 folds, and compute the KMeans silhouette on the held-out fold (mean across folds). Across the nine dense plateau steps in $[48, 80]$, the held-out silhouette is 0.63–0.73 with held-out null mean 0.37–0.42 (gap +0.25 to +0.34); under a 500-shuffle permutation null using the same held-out pipeline, *all nine steps reach* $p < 0.05$ (best $p < 10^{-3}$ at steps 56/68/72), which is stronger than the 7/9 achieved by in-sample selection. The plateau is therefore not an artifact of label-driven feature selection.

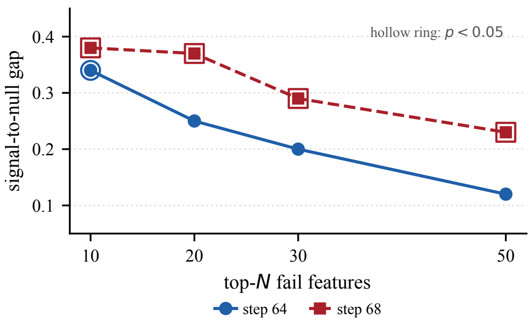

Figure 9: Signal-to-null gap as the top-N fail-feature cutoff is varied at LLaDA-MBPP plateau steps 64 and 68. Hollow rings mark $p < 0.05$. Step 68 is robust across N; step 64 only reaches significance at N=10.

L Activation-Maximizing Examples

To turn the statistically discovered top features into interpretable failure markers, we re-run LLaDA-8B at step 64 on all 257 MBPP samples, encode every generation position through the L26 Mask-SAE, and record the top-30 *(sample, position)* entries per target feature. Table 4 summarises the dominant context type for each feature, the fail-rate in its top-30 examples, and a representative quotation.

feature	dominant context (F/P split)	top quotation
`f15601`	complex-math docstring (25/5)	*"nth octagonal number...O(n)=(n^2+3n)//2"*
`f3892`	`:param`/`:return` annotation (15/15)	"`:param tuple_list:` *input list of tuples"*
`f5561`	arithmetic variable assignment (22/8)	*"max_product = min_product"*
`f11265`	broken-syntax / incomplete code (17/13)	*"dp[n == 0] = [1 \n dp = [0]"*
`f2144`	docstring opening newline (14/16)	*"def fn(x):\n """\n"*

Table 4: Activation-maximising contexts for the top fail-enriched features on LLaDA-8B / MBPP at step 64; fail/pass counts are among each feature's top-30 examples. `f15601` fires on complex-math docstring contexts (task-difficulty correlate), `f11265` fires on broken-syntax contexts (closer to a true failure detector), and `f2144` fires on a structural position (non-discriminative).

Two interpretive points follow. First, the mid-plateau peak feature `f15601` is a *task-difficulty correlate*: its top contexts are dense mathematical specifications (Eulerian / octagonal / hexagonal / Babylonian formulae) where the model is more likely to fail, but the feature is not directly causal. This is consistent with the within-plateau steering null (§4): ablating a task-difficulty signal does not flip correctness because the perturbation is upstream of the failure mechanism. Second, the late-state feature `f11265` fires on contexts that already contain visible syntactic defects in the partially-generated code, suggesting it is closer to a genuine failure detector than the mid-plateau atoms; the within-plateau hand-off from `f15601` to `f3892` (and the late shift toward `f11265`) thus reflects a transition from *difficulty signalling* to *outcome signalling* as denoising completes.

M Functional Correctness Definitions

Every task uses a deterministic, machine-checkable rubric; samples with no extractable candidate are labelled fail. **MBPP:** extract the first `def`-block or fenced "'`python` block, prepend `test_imports`, run all assertions with a 10-second timeout; syntax errors, timeouts, and assertion failures are fails.

JSON schema: output must parse as valid JSON and match the reference structure (top-level type, required keys, scalar-leaf types), ignoring key order and extra keys. **GSM8K:** compare only the final numeric answer; extract `####\s*([-\d.]+)` and compare to gold after stripping commas and trailing zeros. **ARC-Challenge:** extract `####\s*([A-D])`; fallback to the last standalone A–D letter.